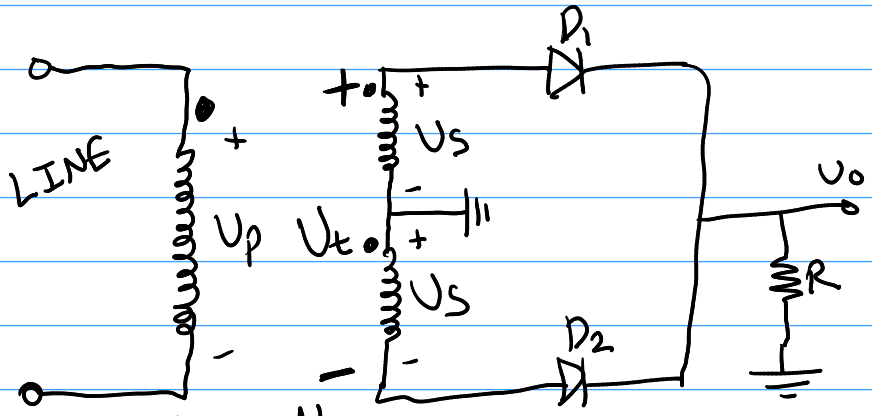
Full Wave Rectification

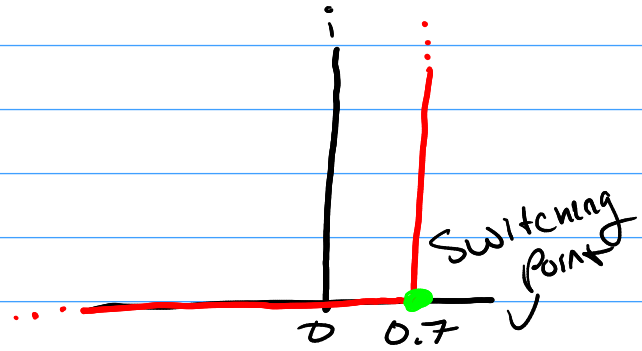
Center-tapped Transformer w/ 2 diodes



$N_p : N_t$

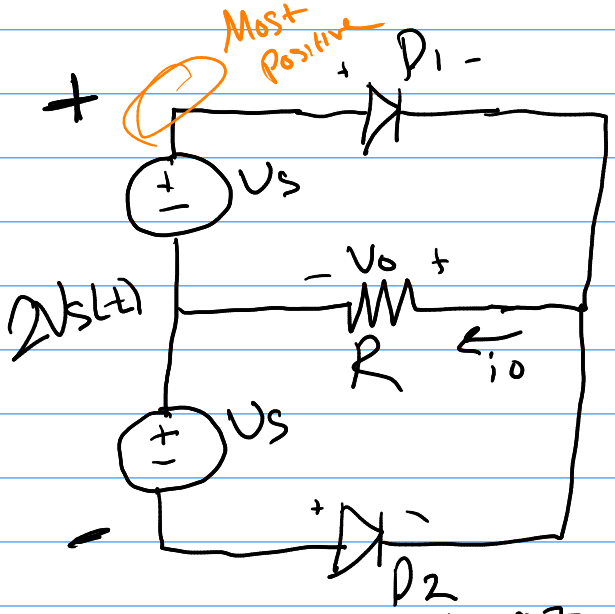
$N_s = \frac{1}{2} N_t$

Use CVD7 Model



for both D's

$\Downarrow$

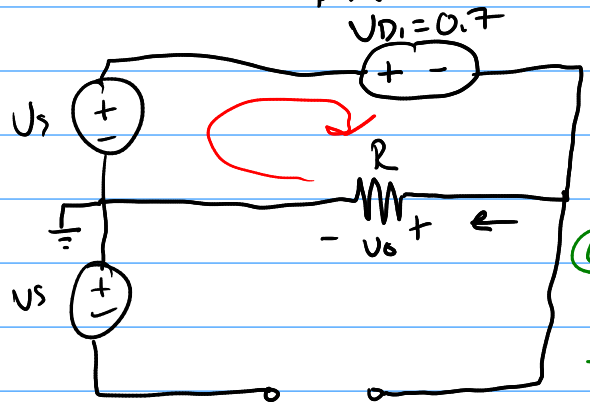


$V_s(t)$ is Sinusoidal

Switching point: $i = 0\,A$
$v = 0.7V$

When $V_s(t)$ (or $2V_s(t)$) is positive enough

D_1 is ON and D_2 is OFF



KVL: through R

$$-V_s(t) + 0.7 + \underbrace{Ri_o}_{V_o} = 0$$

@Switching point: $i_o = 0$ and $V_{D_1} = 0.7$

$$-V_s(t_1) + 0.7 + 0 = 0 \Rightarrow V_s(t_1) = 0.7$$

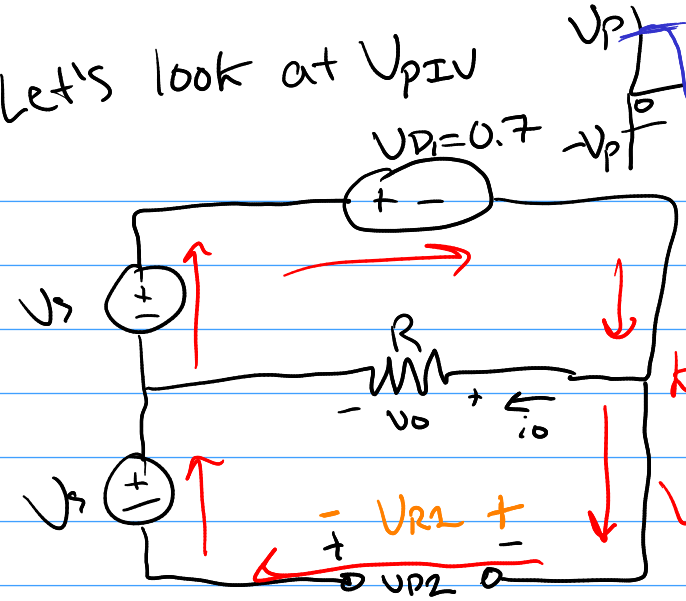
$$\Rightarrow -V_p \cos(2\pi f t_1) + 0.7 = 0$$

$$t_1 = \frac{1}{2\pi f} \cos^{-1}\left(\frac{0.7}{V_p}\right)$$

* Make sure this is in radians

... whenever $V_s(t) > +0.7$ volts, D_1 is ON

Let's look at V_{PIV}



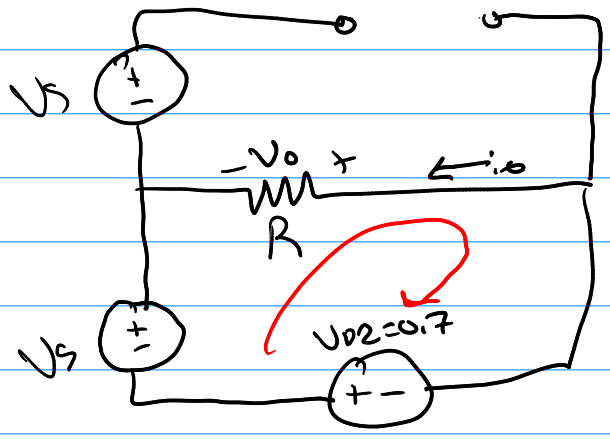
write a KVL that includes D_2 where V_{R2} is reverse voltage for D_2.

KVL: $-2V_s(t) + 0.7 + V_{R2} = 0$

$V_{R2} = 2V_s(t) - 0.7$

$V_{PIV} = \max\{V_{R2}\} = \underbrace{2V_{s_{peak}}} - 0.7$

Suppose $V_s(t)$ is negative "enough"... D_1 is OFF and D_2 is ON



KVL: $-V_s(t) - R\,i_o(t) - 0.7 = 0$

@ Switching point (for D_2)

$V_{D2} = 0.7$ and $i_o(t_2) = 0$

$\Rightarrow -V_s(t_2) - 0 - 0.7 = 0$

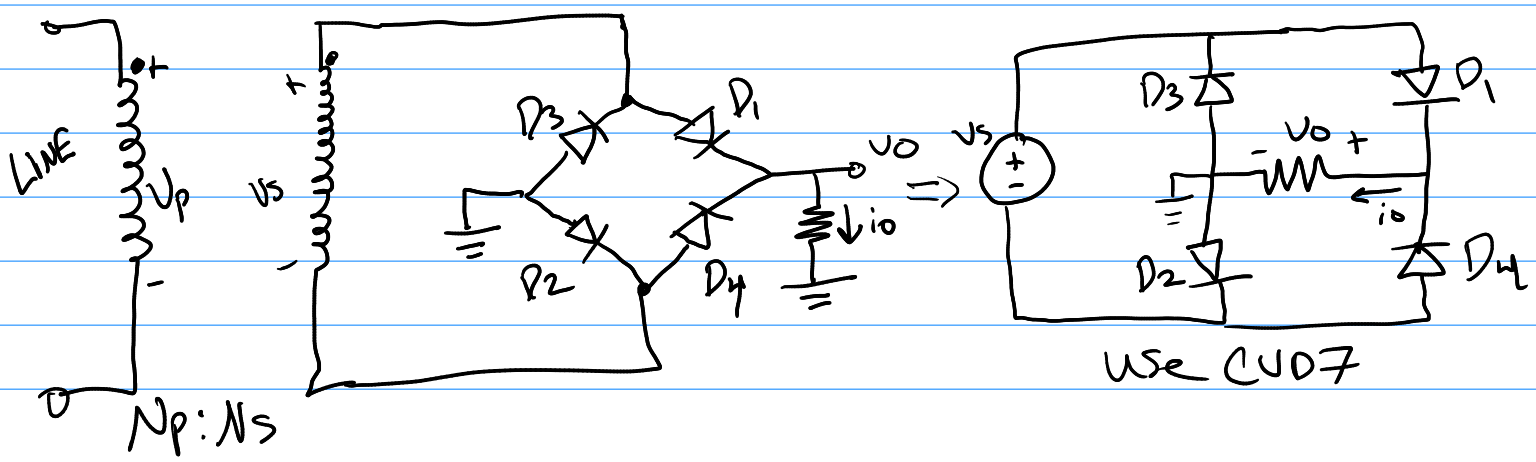
$\Rightarrow V_s(t_2) = -0.7$

D_2 is ON and D_1 is OFF when $V_s(t) < -0.7$

... Would find the same value of V_{PIV} for D_2 as D_1

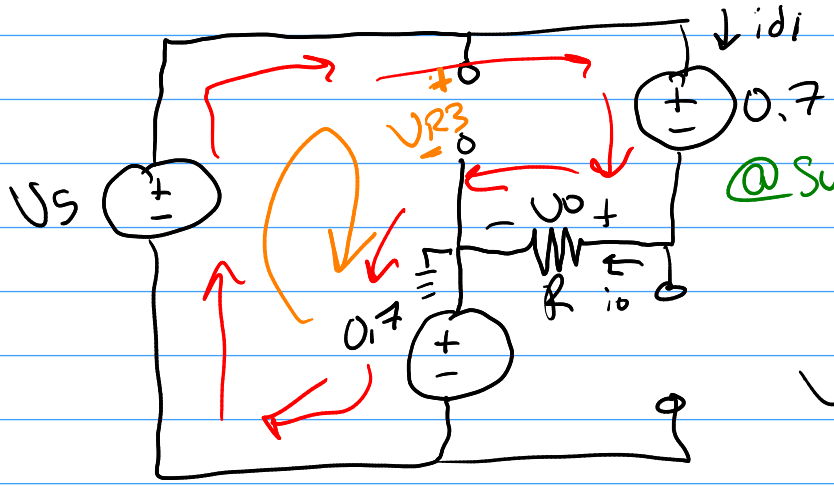
When $-0.7 < V_s(t) < +0.7$, both D_1 and D_2 are OFF

Bridge Rectifier (Full Wave)



use CVD7

When $V_s(t)$ is positive enough D_1 and D_2 are ON
D_3 and D_4 are OFF



KVL: $-V_s(t) + 0.7 + Ri_o + 0.7 = 0$

@ Switching point $\left.\begin{array}{l} V_{D1} = 0.7,\ i_{d1} = 0 \\ V_{D2} = 0.7,\ i_{d2} = 0 \end{array}\right\} i_o(t_1) = 0$

$V_s(t_1) = 1.4$ Volts

Diodes stay the same when $V_s(t) \geq 1.4$ Volts

V_{R3} is $-V_{D3}$

KVL: $-V_s(t) + V_{R3} + 0.7 = 0$

$$V_{R3} = V_s(t) - 0.7$$

$$V_{PIV} = V_{s_{peak}} - 0.7$$

All Diodes are off when
$-1.4 < V_s(t) < +1.4$